2Method and System for Driver's Brain-Computer Interface (BCI)-Based Emergency Takeover Under Autonomous Driving Failure

Abstract: This study focuses on the problem that delayed driver takeover is prone to causing accidents when the autonomous driving system fails, and proposes a BCI-based (Brain-Computer Interface) driver emergency takeover method and system prototype to improve the response speed of driver takeover under autonomous driving system failure and prevent accidents. By adopting the single-channel EEG (electroencephalogram) signal detection device TGAM and UC-win/Road driving simulation software, this study constructed autonomous driving failure scenarios under simulated driving experimental conditions, collected drivers' pre-takeover EEG signals, established a driver takeover intention and mode recognition model and algorithm based on the Long Short-Term Memory (LSTM) neural network, and developed a driver BCI-based emergency takeover system equipment for autonomous driving model vehicles; the experimental results show that the recognition accuracy of this technology for driver takeover intention and mode is approximately 90%. This study innovatively constructed a multimodal driving signal acquisition platform based on a driving simulator, broke through the BCI-based driver takeover intention recognition and rapid takeover technology, and developed a small autonomous driving BCI-based emergency takeover vehicle prototype with real-time performance, fast response, high integration, and modularity; this prototype helps improve the accuracy of driver state monitoring, enhance driving safety, meet the travel needs of people with special requirements, and can be extended to multi-field applications. With the development of brain science and artificial intelligence technology, EEG signal interpretation is expected to become more accurate, which will further improve the reliability of BCI-based emergency takeover.

Keywords: autonomous driving failure, BCI-based control (Brain-Computer Interface), emergency takeover, autonomous driving small vehicle prototype

1.Research Background

Currently, Level 2/Level 3 autonomous driving systems have significantly improved driving comfort and convenience through functions such as Adaptive Cruise Control (ACC) and Lane-Keeping Assist System (LKAS). However, their safety still faces severe challenges in complex scenarios. As autonomous driving technology gradually penetrates real-world road environments, the issue of takeover delays caused by system failures has become increasingly prominent. For instance, in scenarios such as severe weather, road construction, or sudden obstacles, sensors are prone to interference, leading to autonomous driving system failure. At this point, the driver's rapid takeover becomes a core link in ensuring safety. Nevertheless, due to inherent limitations of human physiological mechanisms—such as reaction time, distracted attention, and cognitive load—the traditional manual takeover method carries significant delay risks. Statistics show that the average takeover time for drivers in Level 3 autonomous driving is as long as 2–3 seconds, which is sufficient to cause major accidents in high-speed scenarios.

In recent years, multiple accidents have further confirmed this hidden danger. On August 10, 2022, a Xpeng P7, while driving at high speed with the Lane Centering Control (LCC) function activated, failed to recognize a stationary obstacle due to system issues, and the driver's delayed takeover ultimately resulted in a rear-end collision. On October 9, 2024, when an AITO M9 experienced a sudden system failure, the vehicle exited intelligent driving mode, but the driver failed to take over in time, leading to a severe head-on collision. On March 29, 2025, in an intelligent driving accident involving a Xiaomi SU7, the driver took over the vehicle two seconds before the collision; however, the time was too short to make correct takeover operations promptly, causing the vehicle to deviate from the lane and result in an accident. These cases collectively reveal that autonomous driving systems still have flaws in complex scenarios, as well as the fatal shortcoming of delayed takeover in human-machine collaboration.

Meanwhile, Brain-Computer Interface (BCI) technology, as an emerging and highly promising interactive technology, is gradually gaining prominence in multiple scientific fields. Within the research scope of intelligent assisted driving systems, BCI technology has demonstrated remarkable application prospects. The team led by Professor Wolpaw at the Wadsworth Center in the United States has proven the practical application value of EEG Sensorimotor Rhythm (SMR) in BCI-based communication and control, laying a solid scientific foundation for the application of BCI in intelligent assisted driving systems. The Shanghai Center for Brain Science and Brain-Inspired Intelligence has optimized autonomous driving decision-making by monitoring drivers' EEG activities, promoting the initial integration of BCI and intelligent driving. Additionally, the "brain-controlled vehicle" road test project, a collaboration between Nankai University and Great Wall Motors, has achieved the first breakthrough in China for direct human brain control of vehicle acceleration and deceleration.

However, BCI technology still faces challenges in its application to intelligent driving: technically, its stability and reliability need to be improved to adapt to real driving environments; in terms of human-machine interaction, research is required on its impact on drivers' cognitive load and operating experience. Therefore, based on BCI technology, combined with physiological indicator monitoring and improvements to existing achievements, this project conducts research on a rapid takeover system for autonomous driving failure. This effort aims to address complex issues, improve driving safety, promote technological development, and ensure travel safety and convenience.

2.Design Principle

2.1Design Idea

Currently, Level 2/Level 3 autonomous vehicles still require the driver to take over the vehicle in a timely manner in emergency situations or when exiting the Operational Design Domain (ODD) to ensure driving safety. However, if the driver fails to take over in time, it may lead to traffic accidents. To address the issue of excessively long reaction time when drivers take over vehicles using traditional hand-foot operation methods, this study explores a rapid takeover solution based on Brain-Computer Interface (BCI) technology.

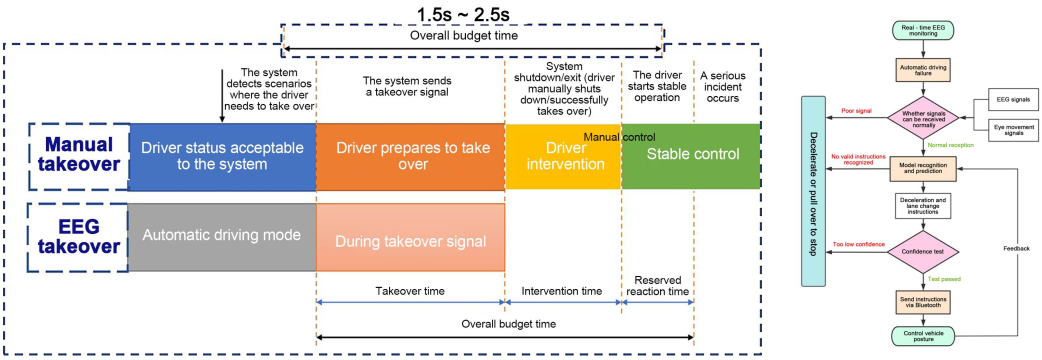

Figure 1: Comparison of Two Takeover Methods and Takeover Flow Chart

This study adopted a single-channel EEG (electroencephalogram) signal detection device and UC-win/Road driving simulation software, and used a driving simulator to construct different takeover scenarios and conduct simulated driving experiments. During the experiment, the study collected drivers’ EEG signals, eye-tracking data, and corresponding takeover commands during emergency takeovers, and constructed a driver command recognition model based on the LSTM (Long Short-Term Memory) neural network. This model enables the rapid and accurate recognition of drivers’ takeover intentions and commands. Based on this model, the study implemented control over vehicle acceleration, deceleration, and steering, realizing rapid BCI-based (Brain-Computer Interface) driver takeover under autonomous driving system failure and effectively reducing the accident probability.

2.2Research Methods

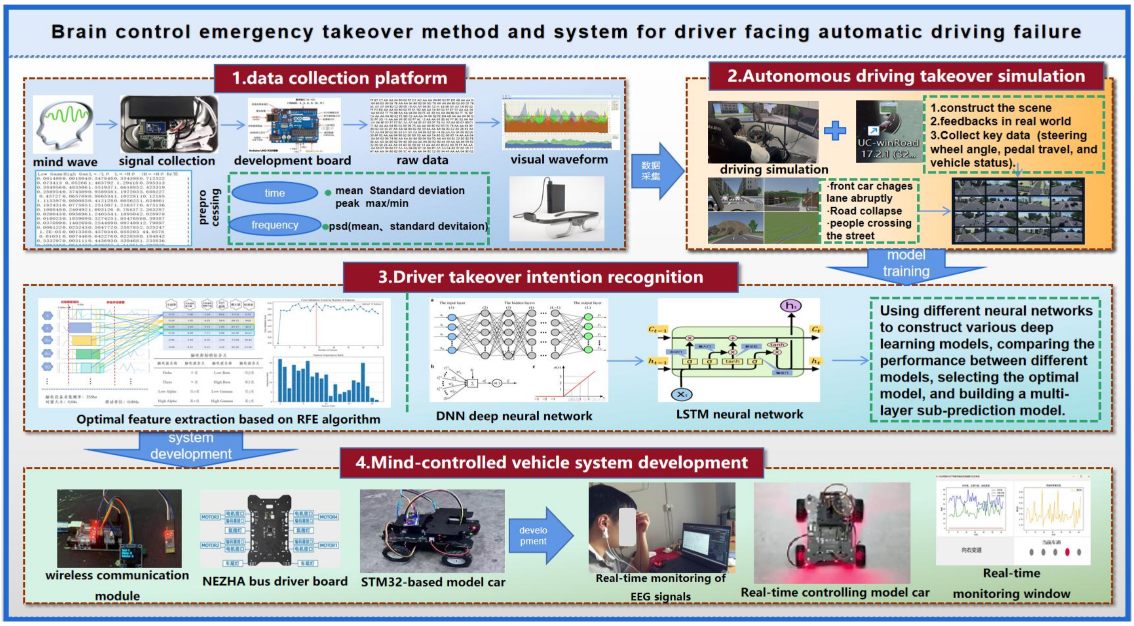

Figure 2: Project Technical Roadmap

2.2.1Construction of Data Acquisition and Preprocessing Platform

a.EEG Signal Data Acquisition

A single-channel EEG device, the TGAM development kit, was used to capture the driver’s EEG activities in real time. The AF3 channel in the frontal region was selected for EEG data acquisition. The frontal region is primarily responsible for the brain’s higher cognitive functions, including decision-making, planning, problem-solving, motor control, emotional expression, and attention regulation, making it suitable for vehicle takeover control. For data acquisition, the HC-05 Bluetooth module was used to receive data packets collected by the TGAM device, which were then processed by an Arduino Uno microcontroller. The Arduino Uno was connected to a computer via a USB data cable to enable power supply, program downloading, and data communication, ensuring real-time transmission and processing of EEG signals.

The Arduino was used to filter and preprocess the received EEG data packets, and key information was transmitted to the computer via a serial port, including Signal (signal strength), Attention (focus level), Meditation (relaxation level), and values of 8 types of EEG Power frequency bands. The preprocessing process ensured the integrity and stability of the data.

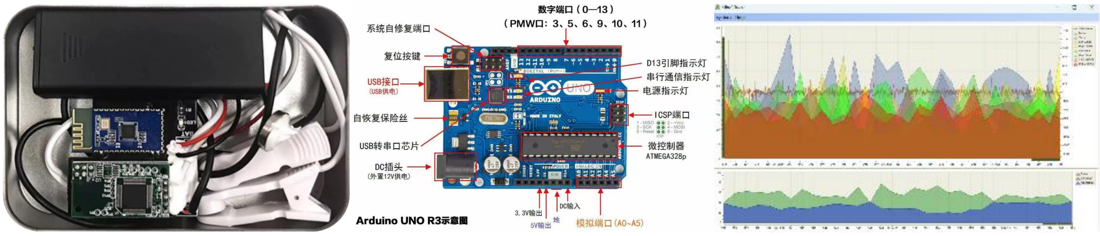

Figure 3: EEG Equipment, ArduinoUNO Development Board and Debugging Interface

Based on this single-channel EEG device, we developed an efficient data processing platform integrating functions of real-time data decoding, signal filtering, and feature extraction. This platform can efficiently analyze the collected raw EEG signals, filter out noise interference, extract key features, and generate visual charts, as specifically shown in Figure 4.

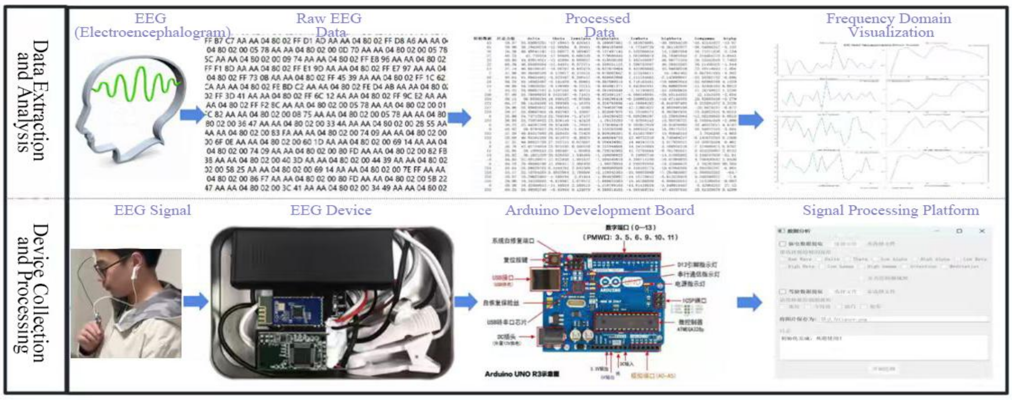

Figure 4: EEG Equipment Signal Acquisition and Data Processing Flow Chart

b.Eye Tracker Data Acquisition

During the simulated driving process, a Tobii eye tracker was used to record the eye movement trajectory characteristics of drivers when processing visual information. The Tobii eye tracker performs eye tracking based on infrared light and camera technology. In data processing, multiple key eye movement features were extracted, including Gaze Point 3D (3D spatial coordinates of the fixation point) and Eye Movement Type (types of eye movement, such as saccade, pursuit, and fixation). These feature data are used to screen out and eliminate interfering takeover segments collected during the experiment, so as to improve the system's accuracy in recognizing driving states and takeover intentions.

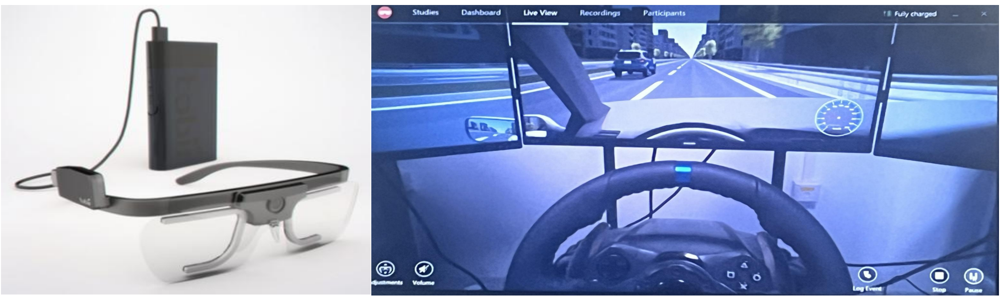

Figure 5: Tobii Eye Tracker and Its Data Acquisition Process

c.Driving Performance Data Acquisition

Vehicle motion characteristics, including timestamps, vehicle 3D position, accelerator, and pedal input values, were collected via the driving simulator. These data were transmitted to the database in real time using the UDP protocol for subsequent analysis and research.

2.2.2Simulation of Takeover under Autonomous Driving Failure

UC-win/Road 17.1.1 driving simulation software was used to construct two types of autonomous driving failure takeover scenarios: simplified scenarios and complex scenarios.

Simplified takeover scenarios include basic operations such as deceleration, left lane change, and right lane change, while complex takeover scenarios cover a variety of driving situations, ranging from simple road construction warnings to sudden events like complex emergency obstacle avoidance. The simulator could accurately reproduce real-world physical feedback and visual presentation to ensure the validity and generalization ability of the research results, providing high-quality input data for subsequent experiments. Road scene diagrams are shown in Figure 6 and Figure 7.

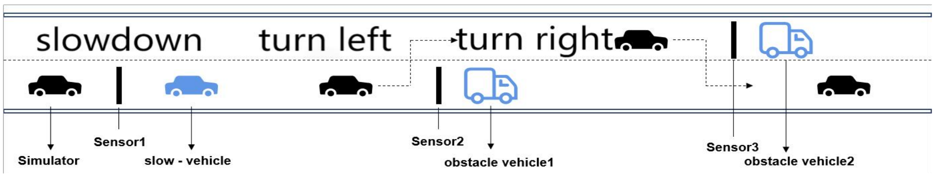

Figure 6: Schematic Diagram of Simplified Road Scenario

Figure 7: Schematic Diagram of Global and Local Views of Complex Road Scenario

Thirty volunteers with different ages, genders, and driving experiences were recruited to conduct a simulated driving experiment. During the experiment, participants were required to respond to sudden situations and perform takeovers while driving. The experiment recorded changes in the drivers' EEG signals during the takeover process, and simultaneously collected key data such as steering wheel angle, pedal travel, and vehicle status. By analyzing the drivers' EEG signals under different takeover states—including left lane change, right lane change, deceleration, and no command issued—and performing data labeling, a comprehensive analysis of the drivers' physiological and behavioral responses during the takeover process was conducted.

2.2.3Driver Takeover Intention Recognition

a.Construction of the Driver Takeover Intention Recognition Indicator System

Not less than 30 types of EEG signal features were collected and extracted, including raw data values, signal strength, attention value, relaxation value, frequency band features (such as Delta, Theta, LowAlpha, HighAlpha, LowBeta, HighBeta, LowGamma, HighGamma, dominant frequency, and mean power spectral density), as well as time-domain features (e.g., mean, variance) calculated with a sliding unit of 0.004 seconds and a time window size of 0.04 seconds. Based on these features, effective recognition of the driver's three types of takeover intentions (deceleration, left lane change, right lane change) was achieved, thus constructing a BCI-based emergency takeover system for drivers.

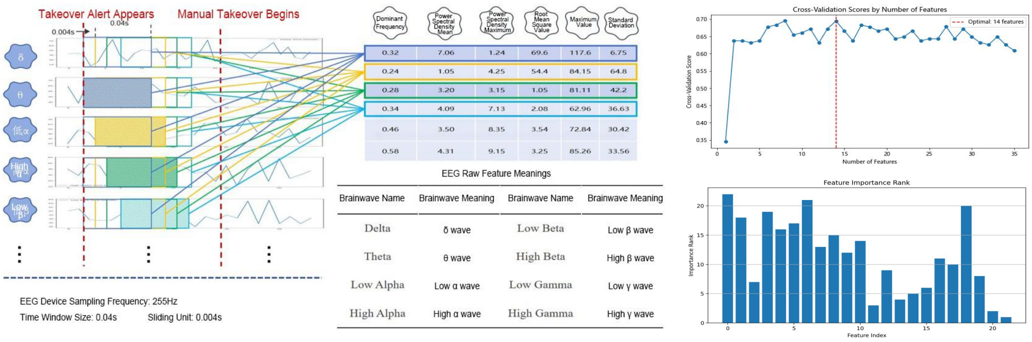

Figure 8: Feature Extraction Methods and Output Result Diagram of Recursive Feature Elimination (RFE) Algorithm

b.Selecting Indicators Based on the Recursive Feature Elimination Algorithm

The optimal indicator combination was extracted from the indicator data based on the RFE algorithm, and the optimal number of indicators and indicator ranking are shown in Figure 5.

It can be seen that the optimal number of indicators for identifying the driver's distraction state is 14. The optimal indicator combination obtained according to the ranking of indicator importance is shown in Table 1. Subsequently, the distraction recognition model will be constructed based on this indicator combination.

Table 1: Optimal Indicator Combination

name	meaning	name	meaning
raw data	EEG raw data	dominant_freq	Main frequency
Delta	δ	psd_mean	Power spectral density mean
Theta	θ	psd_max	Power spectral density maximum
HighAlpha	α high	Rms	Root Mean Square
HighBeta	β high	Max	maximum
LowGamma	γ low	Mean	mean
HighGamma	γ high	Std	standard deviation

c.Construction of the Driver Takeover Intention Recognition Model

Algorithms such as deep learning and reinforcement learning were used to construct a multimodal prediction model for the purpose of achieving accurate prediction of drivers' takeover intentions. The core objective of this model is to identify whether the driver is ready to take over the autonomous vehicle and predict specific control intentions, including deceleration, left lane change, right lane change, and invalid commands.

After constructing multiple prediction models, the distraction state recognition and prediction algorithm based on the LSTM (Long Short-Term Memory) neural network was selected through comparison. The ratio of training set to test set was set at 8:2, and the Adam optimization algorithm was adopted to accelerate the convergence speed and improve the model's accuracy and stability. The EEG signals of drivers when issuing different commands were used as input, and the model outputs the results of driver command recognition (left lane change, right lane change, deceleration, invalid command). The evaluation results of the model are shown in Figure 9 and Table 2.

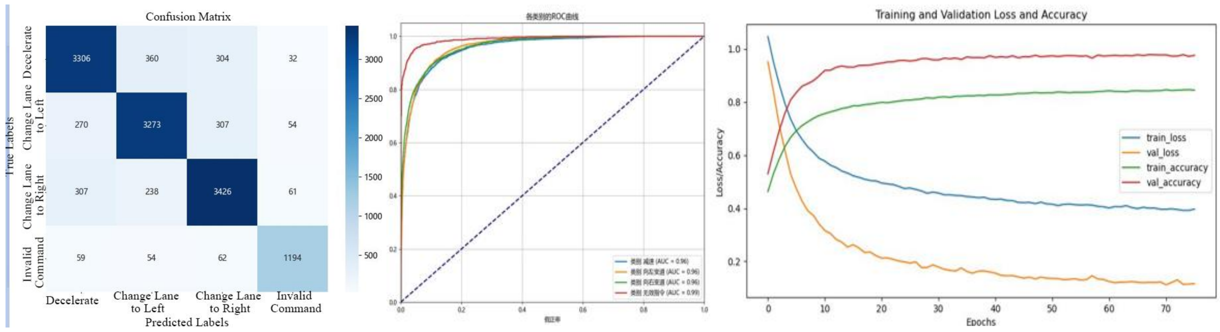

Figure 9: Evaluation Results of the Driver's Takeover Intent Command Recognition Model

Table 2: Output Results of the Driver's Takeover Intent Command Recognition Model

Command	Accuracy	Precision	Recall	Specificity	
Decelerate	0.9561	0.8387	0.8261	0.8323	0.9316
Change Lane to Left	0.8926	0.8339	0.8383	0.8361	0.9307
Change Lane to Right	0.8821	0.8358	0.8497	0.8427	0.9274
Invalid Command	0.9056	0.8904	0.8722	0.8812	0.9877

2.2.4Development of the Driver BCI-Based Vehicle-Mounted System

This study used the STM32-NeZha intelligent car as the model vehicle to construct the driver's emergency takeover system. For the hardware module, the STM32F103C8T6 was selected as the main control unit, equipped with the HC-05 Bluetooth module and the NeZha bus driver board to achieve data transmission and control of the car. The schematic diagram of the driver's BCI-based vehicle-mounted system is shown in Figure 10.

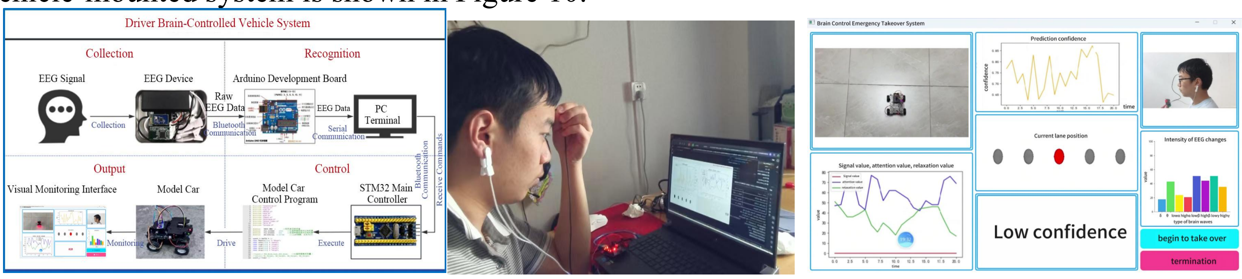

Figure 10: Schematic Diagram of the Driver's Brain-Controlled Vehicle-Mounted System

The system can successfully realize that after the computer receives and processes the driver’s real-time EEG signals, the signals are parsed and converted into corresponding Bluetooth commands, which control the intelligent car to perform actions such as forward movement, deceleration, left lane change, right lane change, and parking. This ensures that in emergency situations, the driver can realize emergency control of the intelligent car via BCI-based control.

3.Innovative Features

3.1Key Innovative Points of the Project

(1) A synchronous acquisition and processing platform for driver takeover control and physiological signals based on a driving simulator was developed.

A platform capable of synchronously collecting drivers' EEG, eye movement, and vehicle manipulation behavior signals was constructed based on a driving simulator. Based on this, autonomous driving failure takeover scenarios covering accident vehicles and road collapse scenarios were established. A large number of driver takeover experiments under simulated driving environments were conducted, and drivers' EEG, eye movement, and takeover operation signals were synchronously collected and stored. Additionally, an EEG signal frequency band decomposition algorithm was integrated and developed to realize the preprocessing of different EEG feature signals.

(2) A takeover intention recognition method based on drivers' EEG was proposed, and the BCI-based emergency takeover technology and hardware-software system for autonomous driving intelligent cars were developed.

BCI technology was creatively applied to realize real-time monitoring of drivers' brainwaves. Based on the RFE feature importance method, 14 pre-takeover driver EEG feature indicators were selected. An accurate recognition method for drivers' takeover intentions (deceleration, left lane change, and right lane change) based on LSTM optimized by Adam was established. Based on this, the BCI-based emergency takeover technology and hardware-software system for autonomous driving intelligent cars was integrated and developed, realizing the function of rapid BCI-based driver takeover under autonomous driving system failure and enhancing the safety of autonomous vehicles.

3.2 Project Features

(1) Real-time performance and fast responsiveness of takeover intention recognition and takeover control based on drivers' EEG signals are achieved.

The rapid takeover mechanism enabled by BCI technology allows the driver's intention to be recognized and converted into vehicle actions within 50 to 100 milliseconds—a level of real-time performance that traditional manual takeover cannot achieve. The system's high responsiveness design ensures that when facing emergency situations, the driver can take prompt action, effectively avoiding accidents caused by delayed takeover.

(2) Modular Design and Prototype Development of the BCI-Based Emergency Takeover System for Autonomous Driving Intelligent Cars

The research achievements of the project have been integrated into the model vehicle-based driver emergency takeover system, thereby realizing the transformation from theoretical research to practical application. The modular design enables the system to be easily expanded and upgraded, allowing for function enhancement in accordance with technological development and market demands, which maintains the system's advanced nature and competitiveness.

4.Application Prospects

1. Ultra-Fast Emergency Takeover: Address Safety Pain Points and Capture the Multi-Trillion Market

When autonomous driving systems fail, BCI technology can convert drivers' intentions into vehicle control commands within 50–100 milliseconds—more than twice the speed of traditional manual takeover (which takes 200–500 milliseconds on average), effectively reducing accident rates. In line with market realities: the WHO reports 1.35 million annual global traffic fatalities, and NHTSA notes that 1 in 4 U.S. accidents stem from distracted driving. This system can cut property damage and insurance claims. Additionally, China's intelligent driving market reached 289.4 billion yuan in 2022 and is expected to approach 1 trillion yuan by 2025; as a core safety component, it boasts broad market potential.

2. Multi-Physiological Signal Fusion: Fill Technical Gaps and Facilitate Industry Development

Current market-based distraction monitoring systems mostly focus on fatigue driving, with single data sources and low accuracy. This system integrates EEG, EDA, and ECG data to significantly improve takeover accuracy, filling technical gaps. Meanwhile, it supports research on traffic accident causes, provides a basis for formulating intelligent driving safety regulations, and promotes the implementation of driving assistance systems.

3. Inclusive Mobility for People with Disabilities: Break Physical Limitations and Expand Niche Markets

Traditional takeover relies on physical functions, excluding people with disabilities. Through "mind-controlled vehicle operation," this system enables people with disabilities to respond quickly when autonomous driving fails, ensuring safety and expanding their mobility freedom. Aligning with barrier-free transportation needs, it opens up new niche markets in intelligent driving.

4. Cross-Domain Expansion: Reuse Technology and Tap into Multi-Scenario Value

The "brain intent recognition + emergency control" framework is transferable: it ensures emergency takeover of unmanned equipment in the military; enables remote intervention in hazardous operation scenarios; supports the safe operation of brain-controlled rehabilitation devices in the medical field; and enhances the authenticity of driving/flight training in the VR field—maximizing the application value of the technology.

5. Technological Evolution: Leverage Brain Science and AI to Maintain Long-Term Advantages

With the development of brain science and AI, the system's algorithms (e.g., Transformer-based time-series analysis) will improve intent recognition accuracy, and BCI devices will move toward miniaturization and non-invasiveness. In the future, using "EEG + behavior" big data, it will evolve from "passive takeover" to "proactive risk prediction," aligning with intelligent driving trends and solidifying its competitive edge.

1